
TELEMETRY + C2 · V1.0

MissionWeaver GCS

User's Guide

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1 Introduction

The MissionWeaver GCS monitors and controls an aircraft from the ground. A handheld unit, it sits inline between your ground station software (Mission Planner, QGroundControl) and an RFD900 radio. It forwards MAVLink unchanged, so your existing tools work as usual, and reads a passive copy to drive its own interface. An internal RC receiver allows you to fly the aircraft manually.

1.1 What it does

- **Monitoring.** Six pages show link and aircraft state: radio, mission, vehicle, sticks, GPS, and HUD.
- **Two GCS connections.** USB serial or USB-Ethernet (MAVLink UDP).
- **On-device radio config.** Edit and apply SiK parameters, local or over-air, without a laptop.
- **RC receiver setup.** FrSky and ELRS binding, stick-mode check, and ELRS firmware flashing.
- **Two-button control.** Every action uses the two physical buttons.

1.2 What's in the box

The device ships fully assembled, with the RFD900 radio and an RC receiver already installed inside. You never have to open the case.

The box contains:

- MissionWeaver GCS unit
- 2x 915MHz Antennas
- USB-C cable
- Quick-start card

Which RC receiver is installed depends on the model you bought. The [Hardware Overview](#) chapter explains how to identify it.

Confirm all items are present before you leave for the flight line. A missing cable or antenna is easier to fix at the bench than in the field.

1.3 Who this guide is for

This guide is for UAS field operators setting up or running a flight with the MissionWeaver GCS. It assumes you can fly the aircraft and operate your ground station, and does not re-teach material their own manuals already cover. Radio and receiver internals appear only as deep as you need to set them safely.

If you are new to the device, read Chapters 2 to 5 in order, then use the guide as reference.

NOTE

This guide covers hardware and software version **v1.0.0**. Screens and menus may differ slightly on other builds.

2 Hardware Overview

The MissionWeaver GCS is a sealed enclosure with a display and two buttons on the front, USB-C and antenna connectors, and the radio and RC receiver mounted inside.

2.1 Display

The display is a 2.42-inch, 128 by 64 pixel monochrome OLED.

2.2 Buttons

The device has two physical buttons, Left and Right. They are the only controls. The [Two-Button Interface](#) chapter covers how they drive the device.

2.3 Connectors

- **USB-C.** Power and the data connection to your ground station. The device has no battery, so it powers on when you connect the cable.
- **Antennas.** The antennas for the RFD900. Attach the supplied antennas before you power on the device.

WARNING

Plug into a USB-C port on the computer when available. Though not usually required, if radio performance is poor, connect the device to a computer that can supply more power.

2.4 Internal radio and receiver

The RFD900 radio is mounted inside and connects to the 915 MHz antenna connector. An RC receiver is installed as well, either a FrSky or an ExpressLRS (ELRS) unit, determined by the variant you ordered.

To confirm which receiver you have, open Settings, then Receiver, and read the configured RC type.

2.5 External receiver

You can use an external RC receiver in place of the internal one. Connect it to the external SBUS input, then select it in software. See Section 7.1 for the steps.

3 Getting Started

This chapter takes you from connecting the device to confirmed telemetry, both on the screen and in your ground station.

3.1 Connect the hardware

The radio and receiver are already inside, so setup is two steps:

1. Attach the supplied antenna to the antenna connector.

2. Connect the device to your computer with the USB-C cable.

WARNING

Attach the antenna before you connect USB-C. Powering the radio without an antenna can damage it.

3.2 Power on and boot

The device powers on when you connect USB-C. It then probes the internal radio, enters AT mode, reads the radio's settings, and opens the MAVLink stream.

The device remembers the last baud rate that worked and tries it first, so a normal boot connects quickly. If that rate fails, for example after you change the radio configuration, it scans the other rates, which takes longer.



Figure 1: The boot sequence, from splash to waiting for MAVLink.

3.3 Connect your ground station

Once the link is established, you may open your ground station software to control the aircraft. In most cases, your ground station software should automatically detect the MissionWeaver and start the feed, via an emulated network connection. If it does not, or if you'd prefer to use a standard serial connection, see Section 6.

3.4 Confirm telemetry

Confirm the link before you rely on it:

1. Check the link indicator in the screen header.
2. Step through the [monitor pages](#) and confirm live aircraft data.



Figure 2: The radio page: link signal, network ID, and frequency.

You can now monitor the aircraft. To control it, bind the RC receiver; see Section 7.

4 The Two-Button Interface

Two buttons run the entire device, Left and Right, and one rule covers almost every screen: *Left navigates, Right acts, a left hold backs out, a right hold confirms an action, and holding both brings up quick actions*. Learn that and you can work the device without looking anything up.

4.1 The five gestures

A click is a quick press and release. A hold means keeping a button down for about a second. From those two actions the device reads five gestures.

Gesture	What it does
Click L	Navigate: previous item, or previous monitor page
Click R	Act: select, confirm, or next monitor page
Hold L	Back out one level, or open Settings from a monitor page
Hold R	Jump to the relevant settings for the current page, or confirm an action
Hold L + R	Open the quick-actions menu, from any screen

4.2 The hint bar

The buttons keep these roles on most screens, but a few screens reassign them. You do not have to track which. After any press, the hint bar at the foot of the screen shows what Left and Right do at that moment, and auto-hides after 2 seconds.

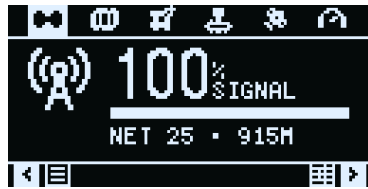
4.3 Confirming risky actions

A few actions taken on the device can drop the link to a flying aircraft, mis-bind the receiver, or push bad settings to the far radio. The device guards each one behind a hold-to-confirm: hold the right button for a few seconds until it commits.

5 Monitoring

Monitoring is the default mode, and it is where you spend most of a flight. The home screen is a ring of six pages. Click Left or Right to step through them, and the ring wraps at both ends. A strip of six icons across the top marks the page you are on, and the header shows “ARM” in the top-right when the aircraft is armed.

5.1 Radio



Link health at a glance: signal strength as a percentage and a bar, the network ID, and the frequency band. Check this page first. When the signal falls, expect telemetry to lag or drop.

5.2 Mission



Progress through the flight plan: the current waypoint out of the total, with the distance and bearing to the next one.

5.3 Vehicle



Power and arm state: battery charge with pack and per-cell voltage, the active flight mode, and how long the aircraft has been armed.

5.4 Controller



The live stick positions, the stick mode and RC type, and the receiver link state. Use it to confirm the transmitter is mapped and connected before you launch.

5.5 GPS



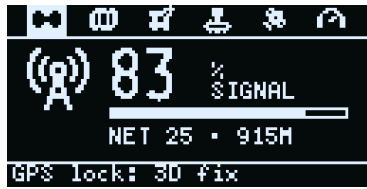
Position and fix quality: satellite count, fix type, HDOP, and latitude and longitude.

5.6 HUD



Attitude and motion: an artificial horizon with altitude, heading, and speed.

5.7 Status messages



The latest message from the autopilot can appear along the bottom of any page, such as a GPS-lock or pre-arm notice. Turn it on under [Display settings](#).

5.8 Auto-cycle

The ring can advance on its own at an interval you set, for hands-free monitoring. It pauses while you press a button and resumes afterward. Enable it under [Display settings](#).

6 Connecting to Groundstation Software

Connect the device to your laptop with the USB-C cable, then open your ground station. Mission Planner and QGroundControl both connect to it automatically. If one does not, add a serial connection by hand.

6.1 It connects automatically

Over USB the device brings up a small network and broadcasts telemetry via UDP. Mission Planner and QGroundControl listen for that stream and connect within a few seconds, with no port or baud rate to set. Watch for telemetry to appear on the [monitor pages](#) and in the ground station at the same time.

The stream is MAVLink over UDP on port 14550, if you need to open a firewall.

6.2 If it does not connect, use serial

The same cable also exposes a standard serial port. Select that port in your ground station and connect. The baud rate does not matter; on a USB connection the setting is ignored.

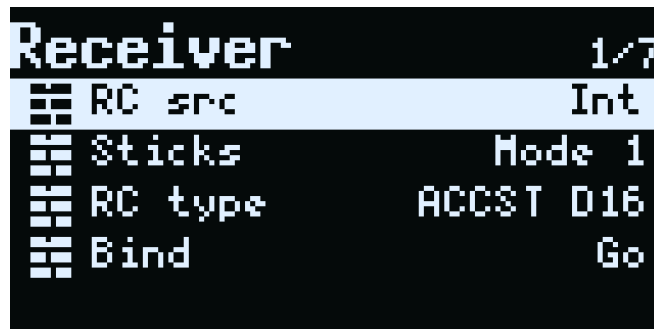
- **Mission Planner:** choose the device's port at the top right, then click Connect.
- **QGroundControl:** open Application Settings, add a Serial link under Comm Links, choose the port, then connect.

7 RC Receiver Setup

The device contains an internal RC receiver for controllers. Binding pairs your controller with the receiver with your transmitter. Set the RC type to match the transmitter, then run the bind wizard. The device drives the receiver and shows each transmitter step on screen.

7.1 Set the RC type and source

Open *Settings*, then *Receiver*. Set *RC type* to match your transmitter: *ELRS*, *FrSky ACCESS*, or *FrSky ACCST D16*. Leave *RC src* on *Int* to use the internal receiver, or switch to *Ext* for an external SBUS input.



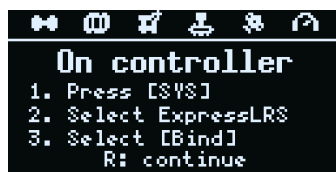
NOTE

Your device ships with the RC type already set for your receiver. Updating with the incorrect software package can misconfigure this, so check this setting after updating if you're unsure.

7.2 Start a bind

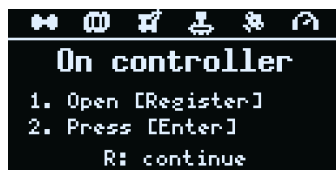
From the *Receiver* menu, select *Bind*. You can also hold Left and Right together from any screen to reach the pair prompt. Choose *Yes* to begin.

7.3 Bind an ELRS receiver



On the controller, press *SYS*, open *ExpressLRS*, then select *Bind*.

7.4 Bind a FrSky ACCESS receiver

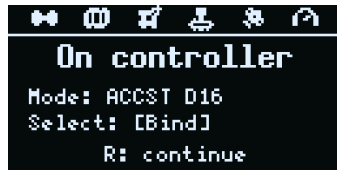


Open *Register* and press *Enter*.

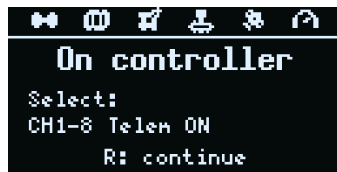


Go to *Receiver 1* and click *Bind*.

7.5 Bind a FrSky ACCST D16 receiver



Set the mode to *ACCST D16* and select *Bind*.



Select *CH1-8 Telem ON*.

7.6 Check the sticks

Open *Settings*, *Receiver*, *Sticks* to watch the live stick positions. If the two sticks do not show the way you hold them, cycle the stick mode, 1 to 4, until the preview matches your transmitter.



The stick mode only changes the on-device preview. It does not change the data sent to the aircraft. Configure stick mixing and channel mapping in your controller's model setup and on the aircraft.

7.7 Flash ELRS firmware

Update the ELRS receiver over its own Wi-Fi portal. When the receiver finds no controller, it starts a Wi-Fi access point you flash from.

1. Power on the device with no paired controller online.
2. Wait 60 seconds. The receiver starts a Wi-Fi network named **ExpressLRS RX**.
3. Join that network from a phone or laptop, then open the portal in a browser.
4. Flash the firmware for the BetaFPV 2.4 GHz RX Lite target.

DANGER

Do not disconnect power during a flash. An interrupted flash can leave the receiver unusable.

8 RFD Radio Configuration

The device has the ability to read and change RFD radio settings on-device, through the *RFD Util* menu.

8.1 Open RFD Util

Open *Settings*, then *RFD Util*. Editing interrupts the live MAVLink link, so the device asks you to confirm before it opens.

WARNING

Editing the radio drops the link to the aircraft. Do not open *RFD Util* while an aircraft depends on that link.

RFD Util		1/27
≡	SerSpd	57600
≡	AirSpd	64
≡	NetID	25
≡	TxPwr	20
⏪		✓

8.2 Edit a parameter

Select a row to change it. Most rows cycle through a list of preset values. Edits are staged, not live. The device holds each pending value against the current one, so nothing reaches the radio until you click apply.

Edit		33
NetID	0..499	
25	x10	
⏪		+

A numeric value such as the network ID opens this editor. Left and Right step the number, and repeated taps grow the step from 1 to 10 to 100.

A link only forms when both radios share the parameters that define the channel: network ID, frequency range, channel count, and air speed. Change one of these and the far radio has to match, which is what *Apply remote* is for.

8.3 Apply your changes

At the end, there are two options to commit changes.

- **Apply** writes the pending values to the local radio. The device reboots, and the link returns on its own.
- **Apply remote** sends the same change over the air to the far radio first, so both ends stay in step.

NOTE

Applying reboots the MissionWeaver. MAVLink flow re-establishes once the device starts back up.



Over-air apply changes the far radio. Hold Right for about 1.2 seconds to confirm, or press Left to cancel.

DANGER

A bad value sent over the air can strand the aircraft's link. Confirm every value before you apply to the remote radio.

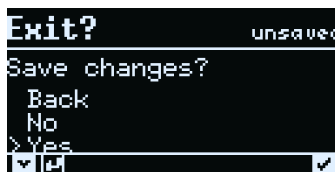
8.4 Re-read, reset, and save

Re-read reloads the current values from the radio and drops any pending edits. *Reset to defaults* restores the radio's factory settings.



Factory reset clears every radio setting. Hold Right to confirm, or press Left to cancel.

If you leave with unsaved edits, the device asks what to do with them.



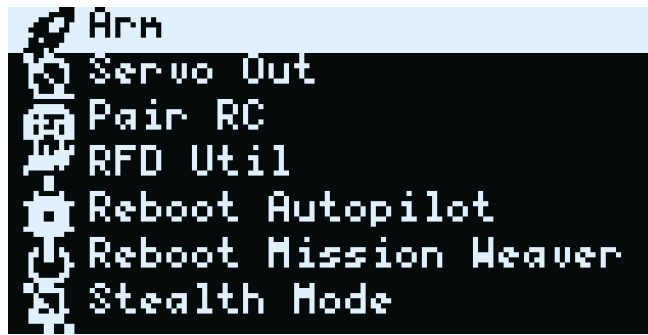
Choose *Yes* to save, *No* to discard, or *Back* to keep editing.

8.5 Configure from a laptop

The Serial interface may also be opened through standard [RFDTools](#), you may connect at any baud rate as the MissionWeaver auto-detects the RFD's baud on startup.

9 Quick Commands

Hold Left and Right together from any screen to open the quick-actions menu. It gives quick access to common commands sent to the aircraft and controls for the device itself.



9.1 Arm and disarm

Arm commands the aircraft to arm, honoring its normal arming checks. Once armed, the same row reads *Disarm*.

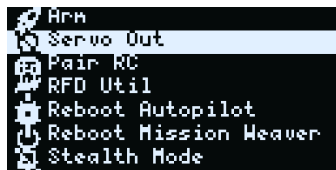


Arming is held behind a confirm. Hold Right for about 1.2 seconds, or press Left to cancel.

DANGER

Arming can spin up the motors. Confirm the area is clear of people before you arm.

9.2 Enable servo outputs

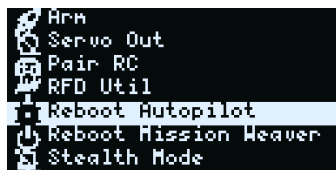


Servo Out turns on the aircraft's outputs, the same as releasing its safety switch. It uses the same hold-to-confirm as arming.

WARNING

Outputs can drive control surfaces and motors. Keep clear before you enable them.

9.3 Reboot the autopilot

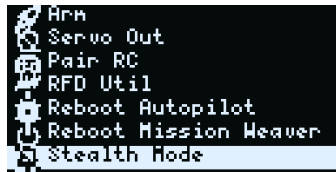


Reboot Autopilot restarts the aircraft's flight controller. The radios stay up, so telemetry returns once the autopilot finishes booting.

WARNING

Rebooting the autopilot cuts flight control while it restarts. Only do it on the ground.

9.4 Stealth mode



Stealth Mode turns off the display and the buzzer, so the device shows and sounds nothing, even through a reboot. Telemetry keeps flowing underneath. Use it for night or low-profile operations.



To leave stealth mode, hold Left and Right together for five seconds.

10 Settings

From a monitor page, hold Left to open Settings. Click Left to move the cursor and click Right to open a category or change a value; hold Left to back out. Changes are saved as soon as you exit the settings menu, so there is no separate save step.



The categories are below, in menu order. *Receiver* and *RFD Util* open the flows documented in their own chapters.

10.1 Display



Brightness and *Sleep*, the idle time before the screen blanks, are self-explanatory. *Cycle* sets how often the monitor ring advances on its own. *Status Text* shows the autopilot's latest message on the [monitor pages](#).



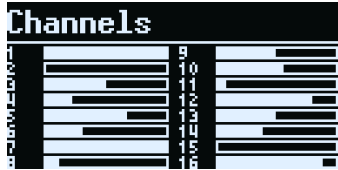
Page+ and *Page-* assign a transmitter channel, 5 to 16, to step the monitor ring forward or back, so you can change pages from a switch on your controller. Channels 1 to 4 are the sticks and cannot be used. Both are off by default.

10.2 Audio

Audio controls the buzzer, the device's only sound. *Buzzer* turns it on or off, and *Volume* sets the level. The buzzer chirps on button presses and warns on blocked or risky actions. It ships off, so the device stays quiet until you enable it.

10.3 Receiver

Receiver sets the RC type and the SBUS source, runs the bind wizard, and reboots the receiver. See Section 7. With an external SBUS source, the receiver-specific rows show *(ext)* and do nothing.



Channels shows the live value of every receiver channel as a bar, so you can check switches, trims, and failsafe before a flight.

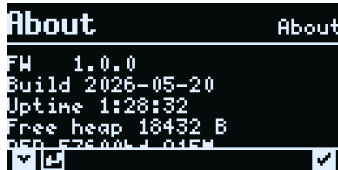
10.4 Vehicle

Vehicle tunes how the device reads the battery. *Cells* is the pack size: leave it on *Auto* to infer the count from resting voltage, or set it yourself when the guess is wrong. *Lo Cell* sets a per-cell low-voltage warning; the buzzer sounds when any cell drops below it. It is off by default.

10.5 RFD Util

RFD Util opens the radio configuration editor, including re-read and factory reset. Entering it drops the MAVLink link, so the device asks you to confirm first. See the [RFD Radio Configuration](#) chapter.

10.6 About

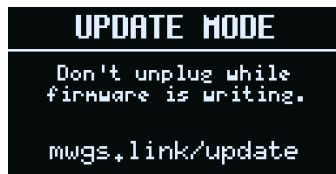


Firmware version and build date, uptime, free memory, and the detected RFD link with its baud and band, or *RFD not detected*. Quote these when you report a problem.

10.7 Update

Update installs new firmware from the device itself. It takes about a minute and needs only the device and its USB-C cable. The firmware ships as a *.uf2* file: in update mode the device mounts on your computer as a USB drive, and copying the file onto it flashes and reboots the device automatically.

1. From a monitor page, hold Left to open Settings.
2. Click Left to move the cursor to *Update*, then click Right to select it.
3. Hold Right to confirm. The device reboots into update mode and appears as a USB drive.
4. In your file explorer, drag the downloaded *.uf2* onto the device's drive. It flashes, then reboots into the new firmware on its own.



Update is guarded by a hold-to-confirm. Hold Right until it commits; the device then reboots into update mode.

Firmware downloads and instructions are at mwgs.link/update.

NOTE

To check the installed version, open *Settings*, then *About*, which lists the firmware version and build date.

DANGER

Do not unplug the device while firmware is writing. The device has no battery, so keep it on USB power until it reboots on its own.

11 Troubleshooting

Most field problems are one of three: no telemetry to the laptop, no radio link, or no control link. Find your symptom below and work the fix.

NOTE

Before blaming the link, check the device's own monitor pages. If they show telemetry, the radio link is good and the problem is on the laptop.

11.1 No telemetry in the ground station

Auto-connect runs over UDP. Mission Planner sometimes fails to release UDP port 14550 from an earlier session, so it cannot pick up the stream and auto-connect errors out. Connect by serial instead, or restart Mission Planner to free the port. See Section 6.

11.2 No radio link, or a poor one



The Radio page reads *LOST*, or a low percentage that will not climb.

A link forms only when both radios agree on the channel: network ID, frequency range, channel count, and air speed. Confirm the local and remote settings match in *RFD Util* or *RFDTools*; see the [RFD Radio Configuration](#) chapter.

On the ground, another RFD radio or MissionWeaver nearby on the same settings can spoil the link. Move away from or power off any conflicting unit. If the link still will not form, reset both RFD900 radios to defaults and set them up again to match.

11.3 No control link



Recover the receiver from *Settings*, *Receiver*, *Reboot RC*.

If the controller is not on and bound within one minute of power-on, the receiver stops looking for a link and starts its own Wi-Fi setup mode. Turn the controller on, then reboot the receiver with *Reboot RC* or power-cycle the device. The receiver searches for the link again on the next start.

11.4 The radio is not detected



Boot stops on *Probing RFD*, *No response*.

The device cannot reach the RFD radio. Check that the radio is seated and powered, then reboot.

11.5 Telemetry is garbled



The device shows *BAUD MISMATCH*.

Bytes are flowing but the device cannot read them as MAVLink, usually because the autopilot's serial speed does not match the radio's. Pick a common rate with Left and Right, then hold Right to set both radios, or set the autopilot to match.

11.6 A button is stuck



The device shows *BUTTON STUCK* and pauses.

A button is held down or jammed. Free it and the device resumes on its own.

11.7 Getting support

Have the *About* page ready if you ask for help. Under *Settings*, *About* it lists the firmware version, build date, and the detected radio.